Category: Cyber security

PS-001: Android Ransomware-Like Activity Early Warning in a Controlled Lab

- **Description:**
  - Signature-based antivirus tools cannot detect zero-day ransomware because the malicious binary has not yet been cataloged. On Android, application sandboxing prevents security software from inspecting other apps' memory directly, but ransomware produces an unmistakable file-system pattern: a rapid sequence of file open, write, and rename operations across local storage. In this project, you will build a behavioral detection tool inside an instrumented Android emulator test environment. By monitoring file-system telemetry within a designated directory, your detector will identify rapid rewriting and renaming patterns, measuring alert delay and file loss across benign workloads and simulated ransomware attacks while terminating only the rogue test workload.
- **Challenge:**
  - Build a behaviour-based early-warning detector for rapid file rewriting and renaming in a controlled Android test environment. The system should identify suspicious activity, generate an alert, and stop only the controlled test workload.
- **Key Requirements**
  - Use one Android emulator or rooted lab device with documented telemetry.
  - Monitor an explicitly accessible test directory.
  - Test with 2 benign workloads and 3 harmless ransomware-like simulations.
  - Measure detection delay, files changed before alert, precision, and recall.
  - Detection must be based on observable behaviour, not real malware.
  - No device-wide prevention claims or monitoring of other apps on stock Android.
- **Expected Outcome**
  - Behaviour-based detection prototype
  - Safe simulation scripts for ransomware-like activity
  - Event/telemetry logs
  - Evaluation results covering detection delay, precision, recall, and files changed
  - Working demonstration in the controlled lab environment
  - Brief documentation of the Android instrumentation and test setup
- **Deliverables***
  1. GitHub Collaboration: (**This step is crucial for allowing mentors to provide real-time oversight.**)
     - Each team must set up a private repository on GitHub.
     - Add our GitHub account (https://github.com/aiori-hackathon) as a collaborator.
     - Once the collaboration request has been accepted all the concerned users (all the teammates as well as aiori-hackathon account) transfer the ownership of the repository to our GitHub account. (https://github.com/aiori-hackathon)
     - Ensure that your repository is named according to your team's name.
  2. Each team is required to upload a PDF to their GitHub repository
     - Please follow the specified structure: Proposed-structure-hackathon.pdf
  3. Prototype Demonstration with a Pseudocode Snippet to the mentor

***All deliverables need to be completed for getting particiaption certificate.**

- **References**
  - https://developer.android.com/privacy-and-security/security-best-practices